Territory Optimization — Impact Report

Savings and benefits from the optimized routing plan.

1. Executive Summary

Per day	Per 2 weeks	Per month
1,445.39 SAR	**17,344.70 SAR**	**34,689.39 SAR**

Across **3** drivers over a **12-workday** cycle, the plan covers **855** store visits. Fleet drives **520 fewer km/day** (600 → 80) and frees **7.1 driver-hours/day**.

2. Savings & Benefits

Item	Per day	Per 2 weeks	Per month	Calculation
Gas saving	130.06 SAR	1,560.70 SAR	3,121.40 SAR	520 km/day × 0.250 SAR/km
Driver time saving	89.14 SAR	1,069.71 SAR	2,139.42 SAR	7.1 h/day × 12.50 SAR/h
Planning time saving	20.83 SAR	250.00 SAR	500.00 SAR	1 h/day × 20.83 SAR/h
Vehicle maintenance savings	0.00 SAR	0.00 SAR	0.00 SAR	Listed qualitatively — not calculated (fewer km → less wear, deferred capex).
Extra revenue from more visits	1,205.36 SAR	14,464.29 SAR	28,928.57 SAR	+11.2 visits/day × 107.14 SAR/store/day
Consistency benefit	0.00 SAR	0.00 SAR	0.00 SAR	Listed qualitatively — stable routes build customer relationships.
Driver morale / ownership	0.00 SAR	0.00 SAR	0.00 SAR	Listed qualitatively — predictable routes reduce fatigue and turnover.
Total (calculated)	**1,445.39 SAR**	**17,344.70 SAR**	**34,689.39 SAR**	Qualitative items excluded.

Fuel 0.250 SAR/km · Driver 12.50 SAR/h · Planner 20.83 SAR/h.

3. Raw Data

All **855** input shop locations are plotted in **shops_map.html** (interactive Leaflet map).

4. Clustering

Clusters shown with concave-hull boundaries in **clusters_map.html**.

Cluster	Routes (days)	Stops	Total distance
Cluster 1	12	285	240.8 km
Cluster 2	12	285	242.0 km
Cluster 3	12	285	474.4 km

5. Routes

Per-driver per-day routes with OSRM road geometry in **routes_map.html**.

6. Driver Schedule

See **routes.xlsx** — one sheet per driver, one column per day, Google Maps links stacked as rows ($\leq$10 stops per link).